ECE323 Lab – Voltage-Controlled Oscillator

I. Introduction
In many systems, timing circuits are needed not only for synchronous digital design but also to control mixed-signal circuits and interface functions. A few examples of clocked analog circuits include analog-to-digital converters, digital-to-analog converters, switched-capacitor filters, and chopper-stabilized opamps. A single IC may require several different clock frequencies for the analog and digital cells. Since these clocks may not be related by simple integer ratios and created with dividers, voltage-controlled oscillators (VCO) are popular circuits to generate the required frequencies on chip. In this lab, we will analyze, design, and characterize one example of a VCO topology.

II. Objectives
The objectives of this laboratory are
1. Calculate the VCO circuit component values to meet a design specification.
2. Learn the KiCad PCB design flow including LTSpice simulation with a sequential capture and PCB layout plan.
3. Use LTSpice to simulate your captured circuits and verify expected performance and capture schematic integrity.
4. Use Kicad to layout the PCB, verify it meets the schematic, and export the fabrication files to a professional outside PCB fabricator.
5. Develop a schematic and PCB test plan at both the subcircuit and overall circuit levels.
6. Assemble and characterize the VCO PCB prototype for subcircuit performance and output frequency vs. voltage for the overall VCO.

III. Motivation
In this semester's lab projects, we will often consider analog and mixed-signal design techniques. VCO's and phase-locked loops (PLL's) provide a great entry point into circuits that incorporate both analog and digital circuit techniques to create a high performance mixed-signal function.

IV. Prelab
Consider the "roughly" conceptual circuit of Figure 1, where a capacitor is charged and discharged thru two switches using complementary clocks.

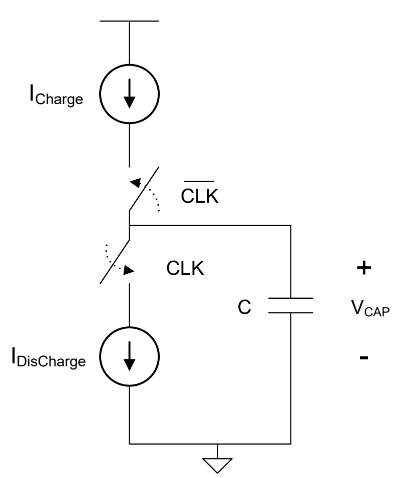


Figure 1. Conceptual Charge/Discharge Circuit

For the case of the charge and discharge currents being equal and the clock signal being 50/50 duty cycle, the capacitor waveform would be as shown in Figure 2.

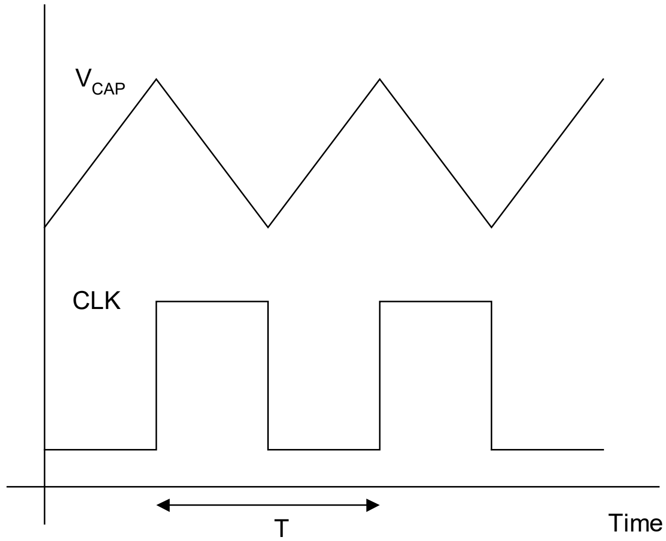


Figure 2. Conceptual Circuit Waveforms

For first half cycle, the top current source will supply (or push) current to the cap and it will charge. For the second half cycle, the lower current source will sink (or pull) current from the cap and it will discharge. The peak-to-peak amplitude of the capacitor transient voltage is then given by (1).

$$\Delta V_{CAP} = \frac{I}{C} \cdot \frac{T}{2} \tag{1}$$

Consider an example embodiment of this idea in the circuit in Figure 3 and description below.

(1) The devices MPCS and MNCS will have respective DC gate biasing (V_{BiasP} and V_{BiasN} from current mirroring configurations) to give a known current flow (I) when operated in saturation. Complimentary MOS devices MPSW and MNSW are driven with a single logic signal $V_{DisCharge}$ with a digital value of either 0 volts (Lo) or V_{CC} volts (Hi) to exclusively turn on either of the switches MPSW or MNSW.

(2) *Charge State ($V_{DisCharge}$ logic Lo):* When $V_{Discharge}$ is logic Lo (0V), the PMOS switch MPSW will be on (triode) and NMOS switch MNSW will be off (cutoff). Current source MPCS will be in saturation and have current flow (I) as long as this device remains in saturation. Current source MNSW will have zero current and zero V_{DS} since the NMOS switch is off. So net result is that capacitor charges.

(3) *Discharge State ($V_{DisCharge}$ logic Hi):* When $V_{Discharge}$ is logic Hi (V_{CC}), the NMOS switch MNSW will be on (triode) and PMOS switch MPSW will be off (cutoff). Current source MNCS will be in saturation and have current flow (I) as long as this device remains in saturation. Current source MPSW will have zero current and zero V_{SD} since the PMOS switch is off. So net result is that capacitor discharges.

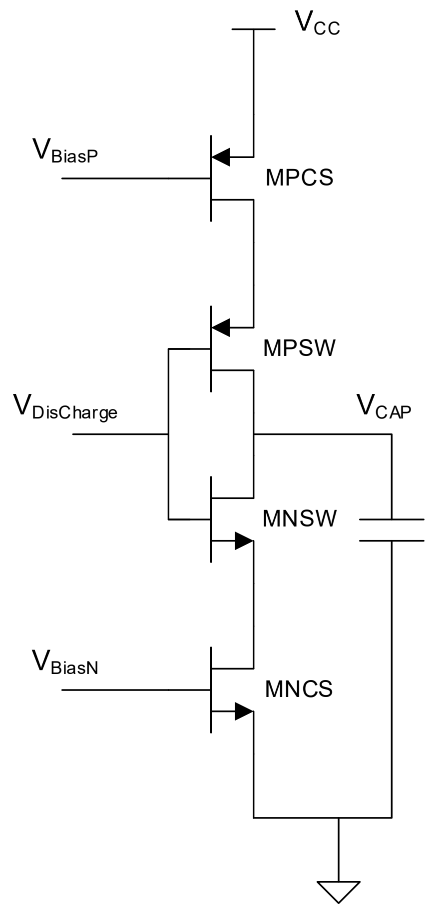


Figure 3. CMOS embodiment of charge/discharge circuit

Now review the VCO circuit of Figure 4 noting the following important functional cells:

1. Voltage to current converter – you should recognize it from last semester. It was used to set the laser current in your audio transmitter board.
2. Current mirrors – we've discussed these in class and they establish the needed gate bias voltages and current sources to create our target charge/discharge currents.
3. Two MOS devices used as switches as discussed above.
4. Integrating Capacitor
5. Voltage divider network to set reference levels for the comparators.
6. Analog Comparators – we will discuss.
7. RS-Latch from NAND gates. (Latch has memory, see Appendix I)

In this arrangement (and if properly designed), the capacitor is unable to attain a DC steady state condition. Notice that the switches in the current steering network are being driven by the output of the latch such that current mirrors will be either pushing current to or pulling current from the capacitor based on the output (FB) logic state. The capacitor voltage is monitored by a window comparator network formed by the analog comparators and the voltage divider and their results fed to an RS-latch. An important architectural feature in this system is that the initial current reference for the current mirror steering network is developed from a voltage-to-current converter. So the input voltage VCTRL sets the oscillation frequency, leading to the overall circuit name of voltage-controlled oscillator (VCO).

Theoretical Investigations:

1. Assuming ideal analog comparators feeding the RS latch, zero delay NAND gates, all NMOS mirror devices identical, and all PMOS mirror devices identical, sketch the transient voltage waveform seen across the capacitor. Label important values. Find an expression for the output frequency with the circuit component values left as variables. In your analysis refer to the two analog voltage divider reference levels simply as V_{HI} and V_{LO}.

2. One important issue with oscillators is whether or not they will start; does this design start from all initial conditions? Explain why or why not.

3. What happens if the charge and discharge currents don't match? Show a sketch.

4. What other ways from a circuit component standpoint can the frequency be changed if the input voltage V_{CTRL} is held constant?

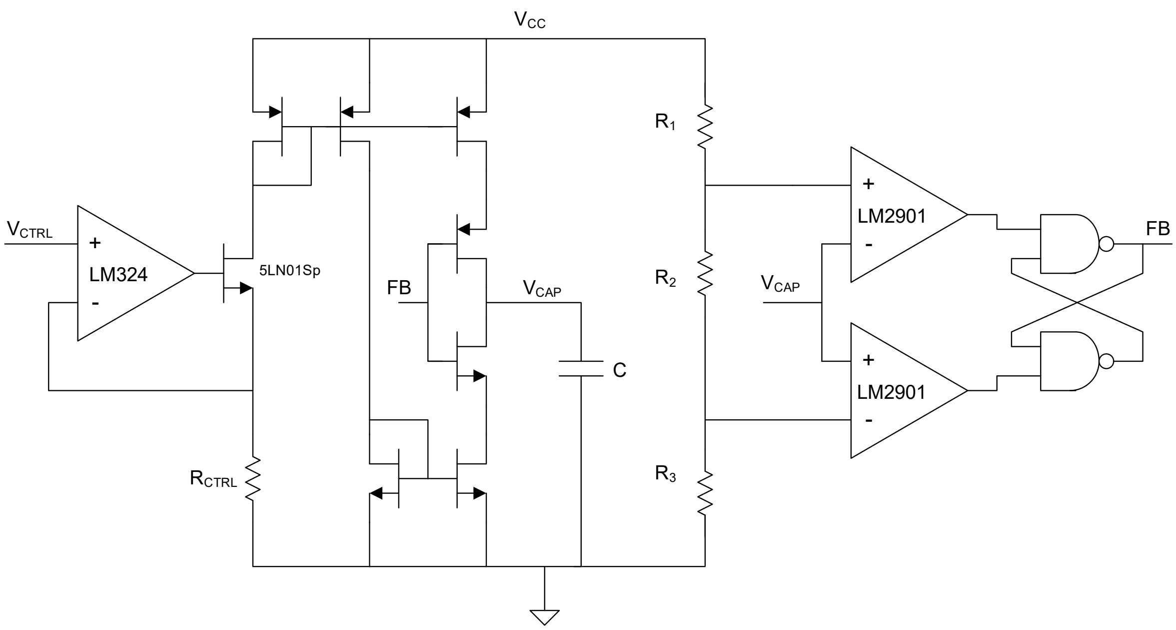


Figure 4. **Conceptual** VCO Topology

V. Additional Circuits

(A) The full design will also include a voltage regulator to set the supply level VCC for the entire board from an input unipolar voltage supply of 10 Volts (VPOWER). The voltage regulator will be an LM317 TO-220 package and we will use the typical design a suggested in the TI LM317 datasheet and shown below.

9.2 Typical Application

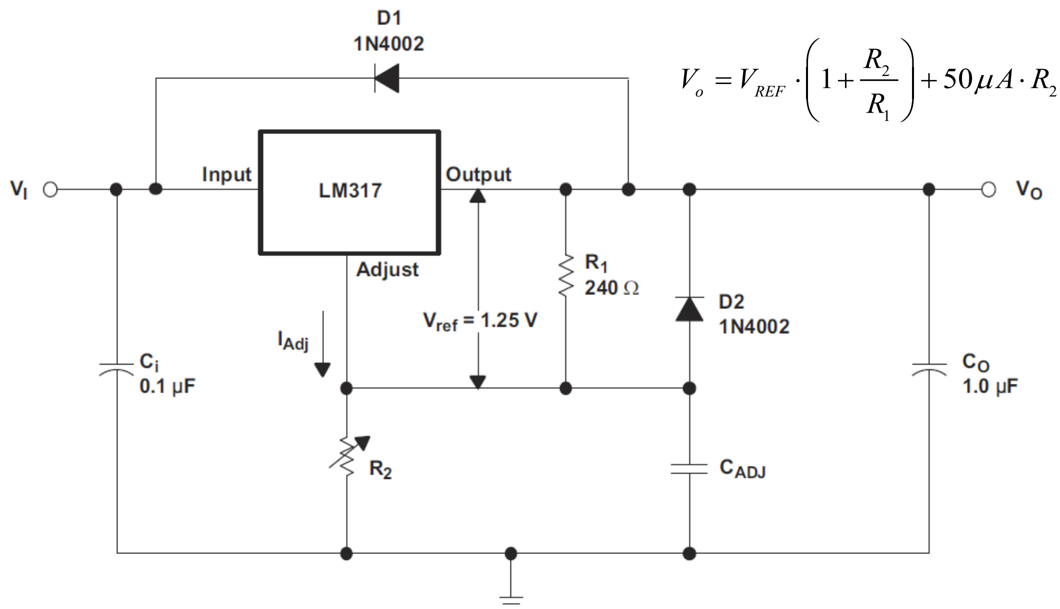


Figure 9. Adjustable Voltage Regulator

Figure 5. LM317 application taken from TI LM317 datasheet

In our case, the regulated VCC voltage (V_O) is to be approximately 5 Volts so R_2 is to be calculated from the relationship in the datasheet. C_{ADJ} is to be a 25uF electrolytic, C_O a 10uF electrolytic, and C_i a 0.1uF ceramic.

(B) The full VCO PCB design shall also include an on-board PWM driven LED circuit for quick verification of PCB functionality.

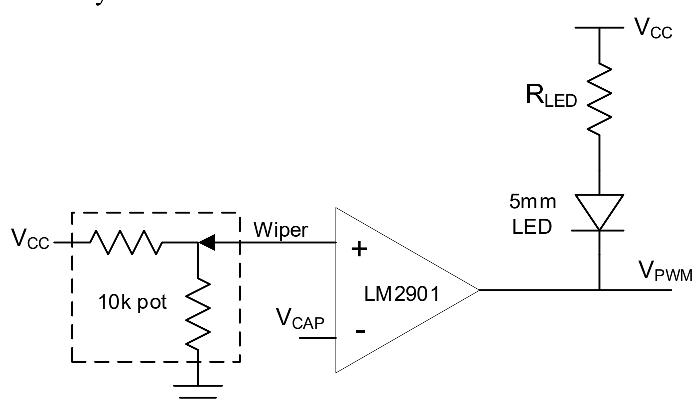


Figure 6. PWM LED driver circuit

(C) The full VCO PCB design will also include a voltage divider on the input VCTRL voltage input line to establish an approximate 1V input when the external VCTRL voltage is not present. The voltage divider should have equivalent resistance input resistance in the low kilo-ohms.

(D) Provide decoupling of IC's and voltage dividers as discussed in the technical notes section.

V. Materials

The materials for this project will include computer workstation for PCB design and simulation. Your bench equipment, fabricated PCB, and provided circuit components for bench verificaiton. Components: LM324, CD4007's, 74AC00, LM2901 (comparators), 5LN01SP, LM317, capacitor, and resistors. Other components needed for the overall circuit will be provided.

VI. Design Specification

You are to design, simulate, build, and test a VCO prototype. In this design, the power supply input is 10 Volts (**no negative supply!**) which is regulated to 5 Volts to supply the entire VCO design, the capacitor value is 0.022uF, and the design should provide an output frequency of approximately 1KHz for an input control voltage of 1.0 Volt.

VII. Technical Notes

1. The LM2901 is a quad comparator IC. Review the device schematic in the datasheet paying attention to the output. This is an open collector device. **You will need to use a pull-up resistor (suggest 10k) from the LM2901 output to VCC (5V) to make the comparator work.** Another issue with the comparator is **input common mode voltage range** which we will discuss in the coming weeks in class. For the comparator to work properly the **comparison voltages you select on the voltage divider should be no larger than 3 volts**.
2. The only discrete transistor is the 5LN01SP in the voltage to current converter (V2I). All other transistors are contained in the CD4007 chips.
3. The CD4007 contains 3 PMOS and 3 NMOS devices that are "partially" isolated. Your project includes the part CD4007 for your schematic capture, you are not to use discrete transistor symbols. This IC has power and ground (**which must be tied to power and ground**) and you will need to figure out how to use 2 of these IC's to design the current mirrors and current steering switches. Unused devices will have some floating terminals. Remember that good mirrors require process matching. Verify the CD4007 symbol pinout with the datasheet.
4. Your RS latch is created from the quad NAND IC 74AC00, see appendix for NAND Latch discussion.
5. IC's should include a 0.1uF decoupling cap between their power and ground pins. You would like these to be near each IC. Ask about this concept.

VIII. Milestones and Deadlines - Preliminary

Milestone 1- (**End of class Wednesday 2/5**) - KiCad tool training and process flow with the LM317 regulator circuit. Implementation of the LM317 voltage regulator circuit. Theoretical design (R_2 calc), schematic capture of the regulator block, LTSpice verification of intended regulator performance, PCB layout and verification, and documented work in a PowerPoint presentation.

Milestone 2 - (**Before class Monday 2/10**) – Overall design of the VCO circuit. Selection of the reference voltages for the window comparator divider and hence the divider resistor values. Calculation of the charge/discharge current for the specified capacitor value and 1kHz frequency spec. Calculation of the V2I converter resistor value for the corresponding 1V VCTRL input at target frequency spec and integrator capacitor value. Schematic capture plan for V2I and current mirror block (For Monday 2/10). Schematic capture plan for window comparator block (For Wednesday 2/12).

Milestone 3 – (**End of Monday 2/10**) – Schematic capture of the V2I and current mirror network, LTSpice verification, PCB layout and verification, and documented work appended to the PowerPoint presentation.

Milestone 4 – (**End of Wednesday 2/12**) – Schematic capture of the window comparator and latch circuitry, LTSpice verification, PCB layout and verification, and documented work appended to the PowerPoint presentation.

Milestone 5 – (**End of Friday 2/14**) – Schematic capture of additional system components and interconnect of the blocks, LTSpice verification, final PCB routing and verification, and documented work appended to the PowerPoint presentation.

Milestone 6 – (**End of Monday 2/17**) – Schematic and PCB cleanup. Export of final layout files to Gerber fabrication database. Upload of Gerber files to JLCPCB. Selection of fabrication options and screen capture of JLC estimate. *Final PowerPoint Update Due by NOON Tuesday 2/18/25*.

Milestone 7 – (**Tuesday 2/18**) – Submission of Gerber files to purchasing for submission of the JLCPCB order.

Milestone 8 – (**TBD, determined by receipt of PCBs**) – Assemble PCBs and bench test. Append bench test results to the PowerPoint document.

*VCO Project Quiz – **Wednesday 2/19/25 (ish)***

IX. Workspace

Each student will work from a shared google drive created by the instructor. Your working KiCad project folder will be located in this drive folder along with all simulation files, documentation, data files, etc. Files are not to reside on windows desktops or to local machine hard drives. They are required to be in this protected space. All simulation will use the Windows version of LTSpice and the text based Spice deck interface.

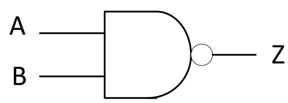
X. Summary

Your lab effort documentation will be a regularly updated PowerPoint presentation including

- Title Slide
- Circuit Schematics (Visio)
- Theoretical Information – How does it work?
- Design Specification and Your Corresponding Design Values with calculations
- Comparison of Experimental and Simulated Results
    - 2 slides of simulated sub-block performance verification (with Visio schematics)
    - 2 slides of measured sub-block performance verification (with Visio schematics)
    - 1 or 2 slides of waveforms, experimental and simulated VCO with 1V on the VCO control voltage.
    - 1 slide with plot of measured and simulated frequency vs. control voltage.
    - 1 slide with discussion of prototype inaccuracy contributors.
- Sequential Circuit and Layout Screen shots from KiCad
- Sequential Verification Simulations
- Conclusions Slide

Appendix I. RS NAND Latch Operation

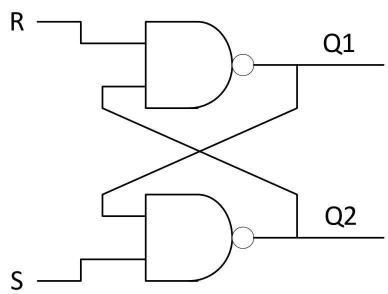
NAND Gate



A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

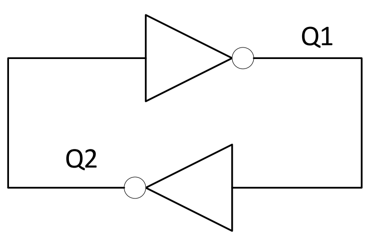
Note: If A=0 or B=0 , Z=1
If A=1, acts as inverter to B and vice versa

RS NAND Latch



R	S	Q1	Q2	
~~0~~	~~0~~	~~1~~	~~1~~	Not Allowed
0	1	1	0	
1	0	0	1	
1	1	Q1*	Q2*	Memory State

Memory Model R=1 and S=1



Positive Feedback!

Q1=1 and Q2=0 is a solution.
Q1=0 and Q2=1 is a solution.
Solution depends on last state, set or reset.
Emphasizes why we don't typically want R=0 S=0 state (we won't know memory state, set by race condition).

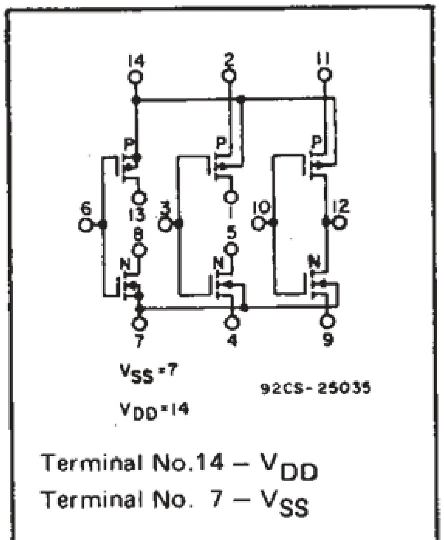
Appendix II. KiCad Reduced Schematic Parts List

1N4001 – Diode
74HC00 Quad Nand
C – Unpolarized capacitor (All caps less than 1uF) Set to standard size ceramics on hand
CD4007UBE – CMOS transistor IC – 3PMOS, 3NMOS 14pin
Conn_01x08 – 8 pin 1 row connector. 2 preplaced on the template schematic for board interfacing
C_polarized_us – Polarized capacitor (caps greater than 1 uF) Set standard size electrolytics on hand
LED – 5mm LED
LM317_TO-220 - LM317 in 3 terminal TO-220 Package
LM324 – Quad opamp 14 pin
LM2901 – Quad analog comparator 14pin
Mountinghold – Preplaced in the template schematic
pwr_flg - Power port identifier for KiCad for schematic checks
Q_NMOS_SDB - Generic 3 pin transistor symbol to instantiate for the 5LN01SP
R – Resistor symbol with package size set to standard 0.25W package
R_potentiometer-US – 3 terminal POT sized selected POT from Digikey

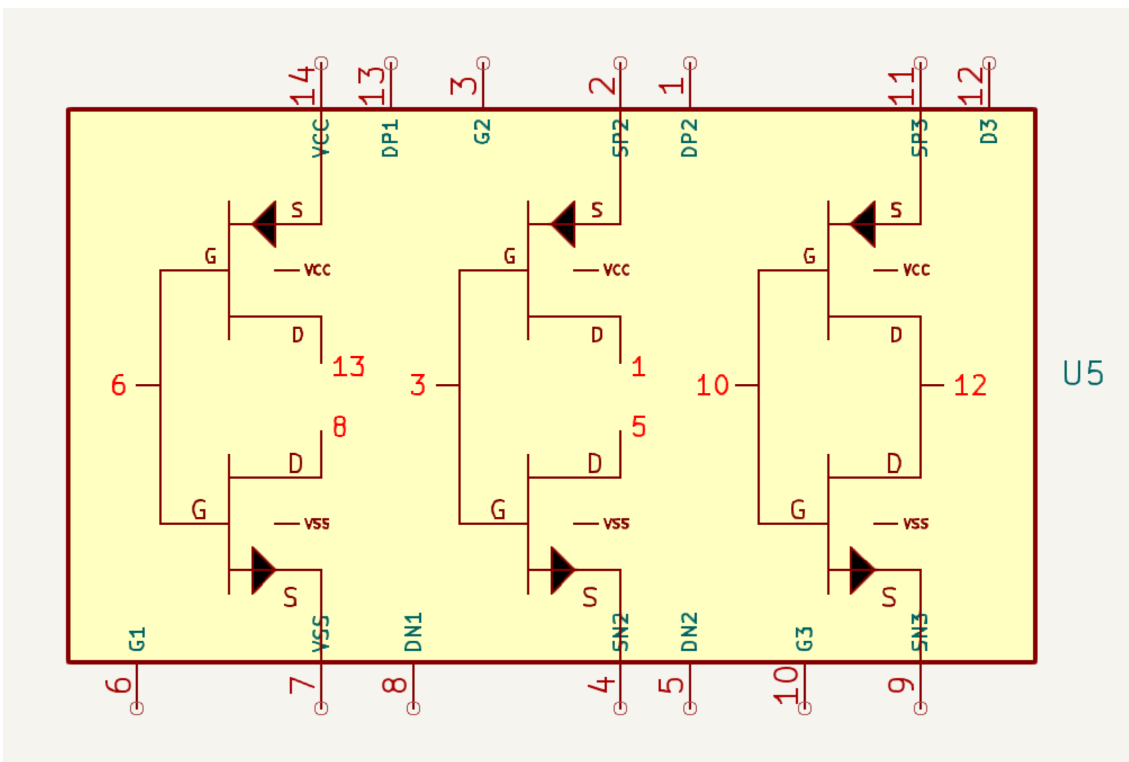
Appendix III. KiCad PCB process flow for the ECE323 (After analysis, component calcs, initial connection maps, etc.)

1. Capture sequential schematic block in KiCad
2. Export text simulation netlist from KiCad
3. LTSpice simulate the full sequential schematic step for correct operation
4. Transfer capture to PCB layout (Update PCB from schematic)
5. Position parts and perform routing
6. Run Design Rules Check which will perform LVS (Layout vs. Schematic connections), clearance, and min/max feature limits.
7. Upon full success, return to top and implement next sequential block.

Appendix IV. CD4007 Functional Drawing from TI and Custom KiCad Symbol



FUNCTIONAL DIAGRAM



Notes: Pin 14 and 7 establish power and ground biasing of all the PMOS and NMOS bodies. So these need to be tied to VCC and RTN.

Appendix V. KiCad Schematic Template

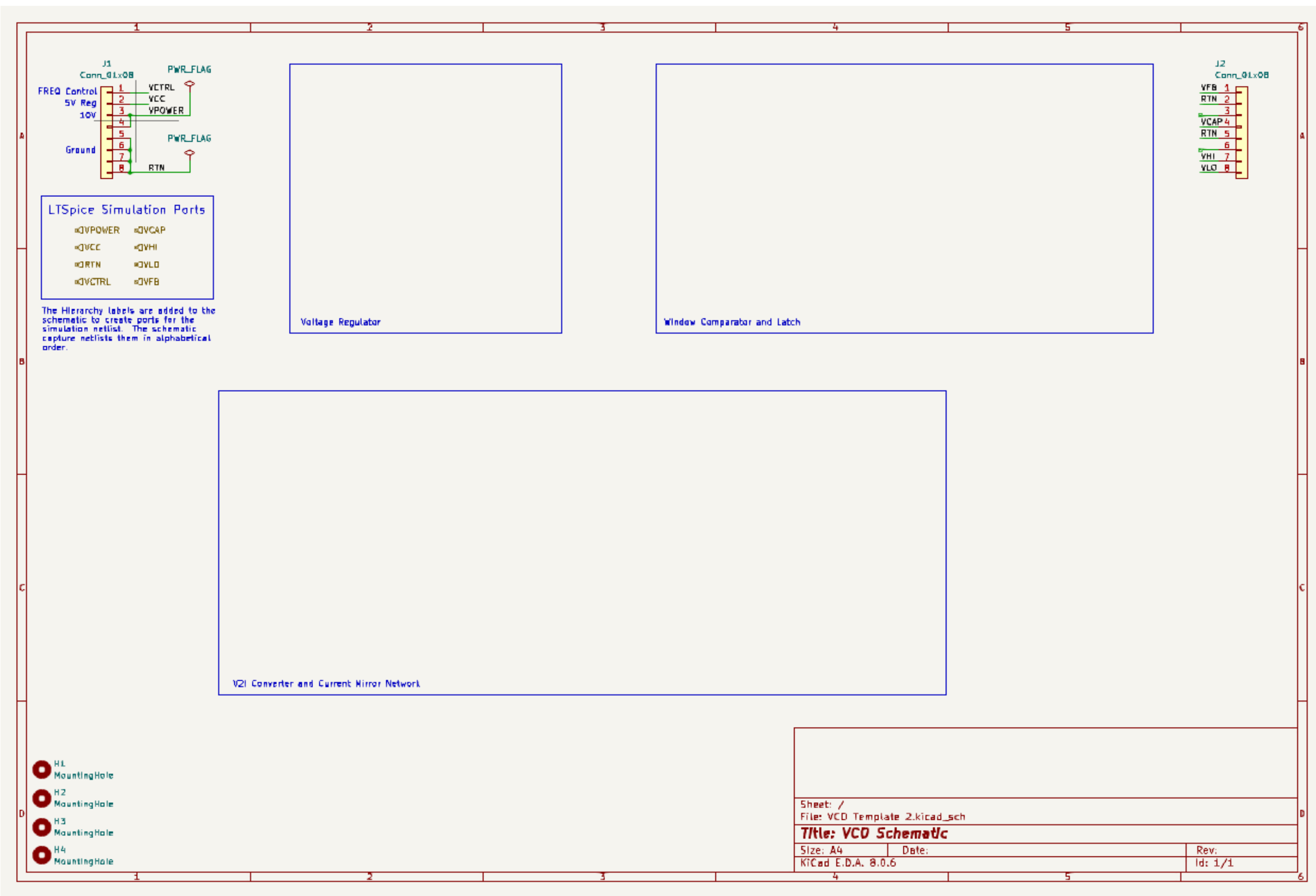
Preplaced J1 connector with pins for interfacing of the VCO control voltage VCTRL, input power and ground VPOWER (10V) and RTN. Preplaced J2 output interface connector with connection to circuit signals VFB, VCAP, VHI, VLO, and RTN for a convenient reference location.

Preplaced power flags for 10V and RTN for Kicad circuit checks.

3 Preplaced boxes for sequential schematic capture effort.

4 Preplaced mounting holes.

Preplaced Hierarchal port names for simulation subcircuit generation (Box labeled LTSpice Simulation Ports). Format of created subcircuit file:
.subcircuit Project_Name
+ RTN
+ VCAP
+ VCC
+ VCTRL
+ VFB
+ VHI
+ VLO



Appendix VI. KiCad Layout Template

1. Predefined board size
2. 2 Preplaced 8 PIN interface connectors
3. 4 Preplace mounting holes
4. Rat lines for RTNs unconnected
5. SilkScreen labels for connectors and VCO Board preplaced.
6. Copper fill regions defined (not obviously seen in this view)